

MONTÉ MAHLUM

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Date of Birth: March 8, 2002
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KEY QUALIFICATIONS

Numerical Methods: Developed, analyzed, and verified numerical methods for PDEs, including proof of convergence for particle methods in nonlinear Wasserstein gradient flow; strong foundations in functional analysis, probability, and mathematical physics.

Machine Learning & Scientific Software: C++, Julia, and Python; re-implemented and validated a graph transformer architecture in Julia against published benchmarks; developed and integrated conflict detection software for air traffic control systems into live test environment.

Research & Collaboration: Lead author of a NASA Technical Memorandum; invited presenter at JMM 2026; experience collaborating across academia, government research, and industry.

PROFESSIONAL EXPERIENCE

Cofounder & Technical Lead, Tapestry RDI

August 2025 – Present

- Developed C++ based software for a Detect and Alert system ensuring safe separation of aircraft; integrated system into a test bench at the NARTP in New Jersey; featured on [CBS Philadelphia](#).
- Deployed Systems Ontology to model architecture of prototype systems, their Test & Evaluation, and their integration with the National Airspace System.
- Leading research program continuing NASA work on a category-theoretic computational framework for systems architecture; two papers in progress.

Mathematical Research Scientist, NASA Langley Research Center

June 2025 – August 2025

- Codesigned domain ontology for systems architecture and developed mathematical and computational framework thereon; worked collaboratively to test framework on SysML-based architecture of the National Airspace System.
- Considered and prototyped various implementations of said framework within the open-source [AlgebraicJulia](#) ecosystem.
- Lead author of resulting NASA [Technical Memorandum](#); subsequent manuscripts in preparation; invited presentation at JMM 2026.

Mathematics Teaching Assistant, University of Minnesota

September 2024 – December 2025

- Led twice weekly recitations and office hours for three courses: Calculus, Linear Algebra, and Differential Equations.
- Instructed 120+ students, emphasizing first-principles mathematical reasoning.

Mathematics Research Assistant, University of Minnesota

July 2023 – August 2025

- Proved convergence of two distinct numerical methods for computing non-linear Wasserstein gradient flow as used in modern machine learning architectures; paper in progress, draft available upon request.

EDUCATION

Master of Science in Mathematics, University of Minnesota *September 2024 – December 2025*
Minor in Computer Science, GPA 3.96 Minneapolis, US

Bachelor of Science in Mathematics, McGill University *September 2020 – June 2024*
Minor in Physics Montreal, CA

Semester Abroad, Univerzita Karlova *February 2023 – June 2023*
Mathematical Physics Prague, CZ

RECENT PROJECTS & PRESENTATIONS

Compositional Systems Architecture (CSA) *Spring 2026*
Prepared comprehensive white paper on a category-theoretic framework for computing and reasoning about assume-guarantee contracts; applications include industrial and biological systems engineering.

Presentation, Joint Mathematics Meeting *January 2026*
For the AMS Special Session on Applied Category Theory, presented the aforementioned work on systems architecture begun at NASA Langley Research Center. Slides made available [here](#).

Re-Implementation of Graph Transformer *Fall 2025*
[Open-source](#) and GPU-accelerated Julia implementation of the [HEAL](#) graph transformer; tightly reproduced results, profiled training on >30k protein structures from the RCSB PDB, and authored a [technical report](#).

TECHNICAL SKILLS

Programming & Computing C++, Python, Julia, Git	Scientific Libraries NumPy, Pandas, PyTorch, SciPy, Matplotlib, AlgebraicJulia, Flow, MJL
Machine Learning Graph neural networks, transformers, modern ML architectures, neural sheaf diffusion	Database Systems PostgreSQL, Neo4j, pgvector
Mathematics & Foundations Category theory, topology & geometry, probabilistic modeling, functional analysis, numerical methods, quantum physics & computation	Systems Architecture & Ontology Ontology engineering, requirements engineering, system design and validation, SysML

LANGUAGES

English – Native	Spanish – Intermediate	Norwegian – Beginner
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INTERESTS

Reading: theoretical physics, mathematical logic, history and philosophy of science, neuroscience, biology. *Writing:* poetry and prose. *Music:* drums, guitar, piano. *Athletics:* langrenn, alpine skiing, slalom waterskiing, rowing, backpacking, hiking, cycling.

REFERENCES

Dr. Ian Levitt, CEO, Tapestry RDI ianl@tapestry-rdi.com
Research supervisor at NASA Langley Research Center, Cofounder at Tapestry RDI

Dr. Li Wang, Associate Professor of Mathematics, University of Minnesota liwang@umn.edu
Research supervisor, academic advisor